Com S 4720/5720 Fall 2025

Problem Set 1 (30 pts)

Due at 11:59pm

Friday, September 5

The first problem is from *https://aimacode.github.io/aima-exercises/*.

1.(20 pts) (Exercise 3.2) Give a complete problem formulation for each of the following. Choose a formulation that is precise enough to be implemented.

(a) (5 pts) There are six glass boxes in a row, each with a lock. Each of the first five boxes holds a key unlocking the next box in line; the last box holds a banana. You have the key to the first box, and you want the banana.

(b) (5 pts) You start with the sequence `ABABAECCEC`, or in general any sequence made from `A`, `B`, `C`, and `E`. You can transform this sequence using the following equalities: `AC` $=$ `E`, `AB` $=$ `BC`, `BB` $=$ `E`, and `E`$x = x$ for any x. For example, `ABBC` can be transformed into `AEC`, and then `AC`, and then `E`. Your goal is to produce the sequence `E`.

(c) (5 pts) There is an $n \times n$ grid of squares, each square initially being either unpainted floor or a bottomless pit. You start standing on an unpainted floor square, and can either paint the square under you or move onto an adjacent unpainted floor square. You want the whole floor painted.

(d) (5 pts) A container ship is in port, loaded high with containers. There are 13 rows of containers, each 13 containers wide and 5 containers tall. You control a crane that can move to any location above the ship, pick up the container under it, and move it onto the dock. You want the ship unloaded.

2.(10 pts) Suppose 8-puzzles have the same goal state S_g shown below.

1	8	3
7		4
2	6	5

S_g

(a) (4 pts) How many inversions are in S_g?

(b) (6 pts) Consider two 8-puzzles respectively with the initial states S_1 and S_2 below. Are these puzzles solvable? Please explain.

1	4	2
	7	3
6	8	5

S_1

3	7	2
6	4	1
5	8	

S_2